# A Multidisciplinary Research Agenda for Artificial Intelligence, Education, Learning, and Instruction

**Abstract**

Artificial Intelligence (AI) is reshaping education, learning, and instruction, yet current research in this area is fragmented, often tool-specific, and dominated by short-term perspectives. This article develops a broader research agenda for AI and Education (AIandED), bringing together Artificial Intelligence in Education (AIED) and AI literacy within an educational ecology framing. Using a collective writing methodology, an expert panel of eleven internationally recognised scholars from various disciplines within computer and learning sciences contributed ten standalone reflections on the challenges, opportunities, and transformations of AIandED. Two additional leading scholars provided critical commentaries to strengthen the analysis. A thematic analysis of the contributions identifies five main challenges (learning and instructional practices and curricula, access and ethics, assessment and evaluation, research capacity, and stakeholder readiness), five areas of opportunity (enhanced pedagogies, innovation in design and research, support for learning processes, critical skills, and hybrid knowledge), and four transformational themes (AI technologies and the design of education, human-AI interplay, lifelong learning, and organisation of AIandED research). The article proposes an educational ecology research agenda across macro (policy, research ecosystem, society), meso (curricula, institutions, leadership), and micro (instructors, learners, learning processes) levels. We argue for

a future-oriented, critical, and inter- or multidisciplinary approach that recognises AI as a socio-technical assemblage and sustains educational values such as equity, democracy, and human dignity in postdigital societies.